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Binding Affinity Predictor

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Executive summary

The long-term objective is a controlled comparison of Boltz-2, Nesso-1, and FlashBind: what each model receives, what quantity it predicts, and how its claims survive a common benchmark. Boltz-2 motivates particular care because cofolded structure confidence and its affinity-related outputs are distinct objects [1].

This update first develops a high-level picture of Boltz-2 and then uses it to explain Nesso-1. Boltz-2 combines a learned representation trunk, full-atom diffusion, confidence estimation, and affinity prediction. Nesso-1 retains a coarse representation of protein–ligand geometry while removing the expensive full-atom decoder, making it primarily a rapid affinity-screening model [2].

1 Scientific questions

1. What molecular information do Boltz-2 and Nesso-1 receive?
2. How does each model represent the protein, ligand, and their possible geometry?
3. Why does Boltz-2 generate full-atom coordinates while Nesso-1 does not?
4. What do their binder and continuous-affinity outputs mean?
5. What physical detail does Nesso-1 sacrifice to obtain speed?

2 Boltz-2 architecture: a high-level view

Boltz-2 combines two related tasks: it first predicts a full-atom biomolecular complex and then uses the predicted interface to estimate whether the ligand binds and how strong that binding may be [1]. The model is easiest to understand as four connected components: a trunk, a diffusion decoder, a confidence module, and an affinity module.

Molecular inputs $\longrightarrow$ representation trunk $\longrightarrow$ full-atom diffusion $\longrightarrow$ candidate 3D complexes
$\longrightarrow$ confidence ranking $\longrightarrow$ binder probability + continuous affinity score

2.1 Inputs and molecular representation

For a protein–ligand example, the basic inputs are the protein sequence and a ligand description such as SMILES. The model can also use a protein multiple-sequence alignment, structural templates, experimental-method labels, and user-provided contact or pocket constraints. An atom-attention encoder combines detailed atomic features into token representations. Protein residues are represented as residue-level tokens, whereas ligand heavy atoms are represented individually.

The trunk maintains two complementary learned objects:

- a *single representation* s_i describing each token; and
- a *pair representation* z_{ij} describing the relationship between every pair of tokens.

Template and MSA information are incorporated before a 64-layer PairFormer updates these representations. Recycling sends the updated representations through the trunk repeatedly so that sequence, molecular identity, and putative geometry can be refined together.

2.2 Full-atom diffusion decoder

The trunk does not itself provide the final coordinates. Boltz-2 begins the structure-generation stage with noisy, effectively random atomic coordinates and applies a learned denoising module repeatedly. At a schematic level,

$$\mathbf{x}_t \longrightarrow \mathbf{x}_{t-1} \longrightarrow \cdots \longrightarrow \mathbf{x}_0,$$

where the trunk representations condition every denoising step. The final $\mathbf{x}_0$ contains explicit coordinates for the modeled atoms in the protein–ligand complex. Because the initial coordinates and sampling process are stochastic, several candidate complexes can be generated from the same input.

Optional steering potentials can bias this reverse-diffusion process toward user constraints or improved physical plausibility. These potentials guide sampling; they do not turn the neural decoder into a molecular-dynamics or free-energy calculation.

2.3 Confidence module

A separate, smaller PairFormer examines the final trunk representation and a predicted coordinate set. It estimates confidence quantities such as local and pairwise errors and interface confidence. These values help rank alternative structure samples. Confidence answers "how trustworthy is this predicted geometry?"; it is not itself a measurement of binding strength.

2.4 Affinity module

The affinity calculation uses both the learned trunk representation and the predicted structure. In the reported inference protocol, five structures are generated and the candidate with the highest protein–ligand interface confidence is selected. A specialized PairFormer then focuses on protein–ligand and intra-ligand pairs, combines z_{ij} with a distogram derived from the selected coordinates, and pools the interface into one global representation.

Two multilayer-perceptron heads produce distinct outputs:

1. a binding-likelihood probability intended for binder/non-binder classification; and
2. a continuous, logarithmic affinity value intended mainly for ranking related ligands against the same target.

The continuous head is trained on a mixture of K_i, K_d, and IC_{50} measurements standardized to micromolar units. It should be interpreted as an IC_{50}-like measure of binding strength, not as an equilibrium constant from one uniform assay and not as a calculated ΔG_{bind}.

2.5 What Boltz-2 gives us—and what it does not

Boltz-2 therefore provides two kinds of predictions that must remain separate:

Output family	**Scientific interpretation**
Structure and confidence	A sampled full-atom complex and the model's estimated uncertainty in that geometry
Binder likelihood	A learned classification probability for interaction
Continuous affinity	A heterogeneous-assay, IC_{50}-like ranking score

The affinity estimate can fail when the model selects the wrong pocket, misplaces the ligand or protein conformation, or omits important context. The released affinity module does not explicitly model contributions from water, ions, essential cofactors, or all binding partners. Full-atom coordinates make mechanistic inspection possible, but they do not guarantee physical validity or thermodynamic accuracy.

3 Nesso-1 architecture: a high-level view

Nesso-1 asks whether binding affinity can be estimated from an approximate map of the protein–ligand interface without first generating every atomic coordinate. It is therefore best understood as a *coarse-grained cofolding affinity model*, rather than as a full-atom structure predictor [2].

Protein sequence + ligand SMILES $\longrightarrow$ protein and ligand embeddings $\longrightarrow$ pairwise distance representation $\longrightarrow$ likely binding-region crop $\longrightarrow$ binder probability + continuous affinity score

3.1 Inputs and molecular tokens

The inputs are a protein amino-acid sequence and a ligand SMILES string. No experimental complex, user-specified pocket, or multiple-sequence alignment is required. The protein receives contextual embeddings from the pretrained ESM-2 protein language model. Each protein residue is represented geometrically by one center, its Cβ atom (Cα for glycine), while each ligand heavy atom remains an individual token. Thus, the protein is strongly coarse-grained but the ligand retains its heavy-atom graph.

3.2 A learned map of pairwise geometry

The central object is a pair representation z. For every pair of tokens, z_{ij} stores a learned description of their relationship. Protein–ligand entries answer questions such as "is residue 42 likely to be close to ligand atom 5?" A streamlined PairFormer updates this matrix using triangle attention and triangle multiplication, operations that allow relationships among three tokens to constrain one another.

The trunk predicts a categorical distance distribution,

$$P(d_{ij} \in \text{distance bin } k),$$

rather than one exact coordinate set. From this distribution, Nesso-1 obtains an expected distance and an entropy that measures uncertainty in the predicted protein–ligand geometry.

3.3 Finding the relevant protein region

The first trunk pass considers the full protein. Subsequent recycling steps retain protein tokens predicted to lie within approximately 22 Å of the ligand. The affinity module uses a tighter crop: residues predicted to lie within approximately 15 Å of any ligand heavy atom. In this sense, Nesso-1 infers a likely interaction region internally rather than requiring the user to provide a pocket.

3.4 Affinity outputs

A second, smaller PairFormer—the affinity module—combines the protein language-model embeddings, ligand features, pair representation z, and its predicted distance information. It produces two distinct outputs:

1. a binder likelihood for hit-identification classification; and
2. a continuous affinity or potency score for compound ranking.

The continuous output is not a calculated ΔG_{bind}. During training, K_i, K_d, IC_{50}, and EC_{50} labels are treated interchangeably to accommodate incomplete and heterogeneous public assay metadata. It should therefore be interpreted as an assay-trained affinity-like score, with particular value for ranking compounds measured in comparable settings.

3.5 The essential contrast with Boltz-2

	Boltz-2	**Nesso-1**
Structural decoder	Full-atom diffusion producing explicit coordinates	No heavy-atom diffusion; predicts coarse pair distances
Protein context	MSA can be used	ESM-2 embeddings; no MSA
Primary structural output	Full biomolecular complex	Coarse interface representation; approximate poses can be reconstructed
Design priority	Detailed structure plus affinity	Rapid affinity screening

Removing full-atom diffusion is the main source of Nesso-1's speed. The cost is that it cannot resolve side-chain conformations, hydrogen-bond geometry, steric clashes, or water-mediated interactions with atomistic precision. Its architectural hypothesis is that coarse interface geometry may be sufficient for rapid affinity ranking, even though full physical detail remains important for mechanistic interpretation and late-stage optimization.

4 Next steps

1. Develop an equally clear architecture summary for FlashBind.
2. Define a common native-output schema without silently equating binder probability, heterogeneous potency labels, and binding free energy.
3. Choose a small benchmark with exact protein constructs, ligand stereochemistry, assay identity, qualifiers, and experimental units.

Reproducibility

The architecture descriptions are grounded in the downloaded version-1 technical reports for Boltz-2 and Nesso-1. The accompanying manifest records the project Git revision, Overleaf revision, citation keys, and compiled PDF checksum.

References

[1] Saro Passaro, Gabriele Corso, Jeremy Wohlwend, Mateo Reveiz, Stephan Thaler, Vignesh Ram Somnath, Noah Getz, Tally Portnoi, Julien Roy, Hannes Stark, David Kwabi-Addo, Dominique Beaini, Tommi Jaakkola, and Regina Barzilay. Boltz-2: Towards accurate and efficient binding affinity prediction. *bioRxiv*, 2025. doi: 10.1101/2025.06.14.659707. URL https://doi.org/10.1101/2025.06.14.659707. Preprint, version 1, posted 18 June 2025.

[2] Nikhil Shenoy, David Errington, Emmanuel Bengio, Kacper Kapuśniak, Kerstin Klaeser, Yui Tik Pang, Vladimir Radenkovic, Prudencio Tossou, Therence Bois, Andrew Wedlake, and Francesco Di Giovanni. Nesso-1: Accelerating open-source binding affinity predictions. *bioRxiv*, 2026. doi: 10.64898/2026.08.01.742196. URL https://doi.org/10.64898/2026.08.01.742196. Preprint, version 1, posted 3 August 2026.